Date: 09/07/2026

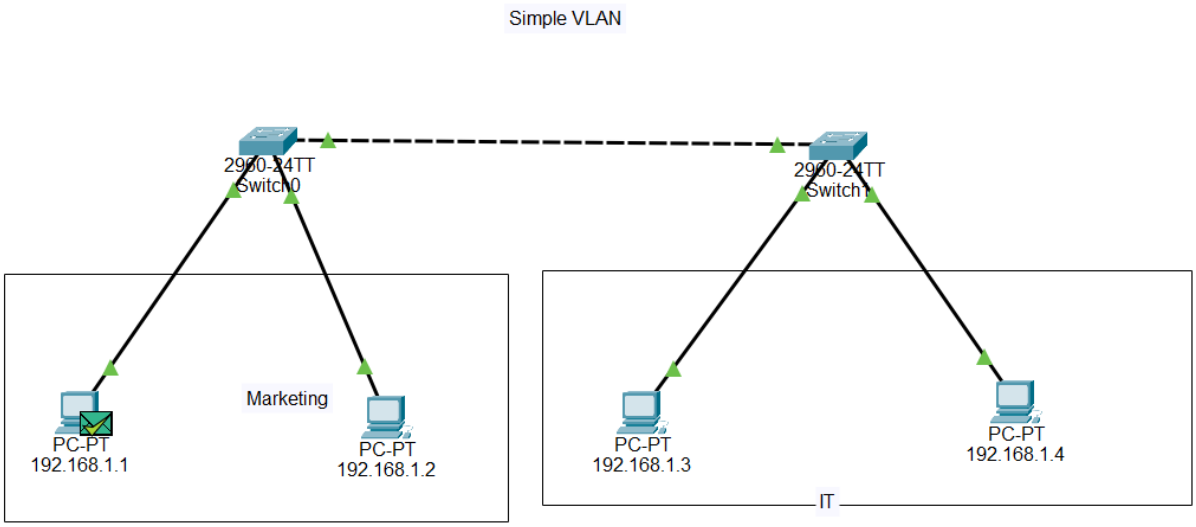
Lab Practical #05:

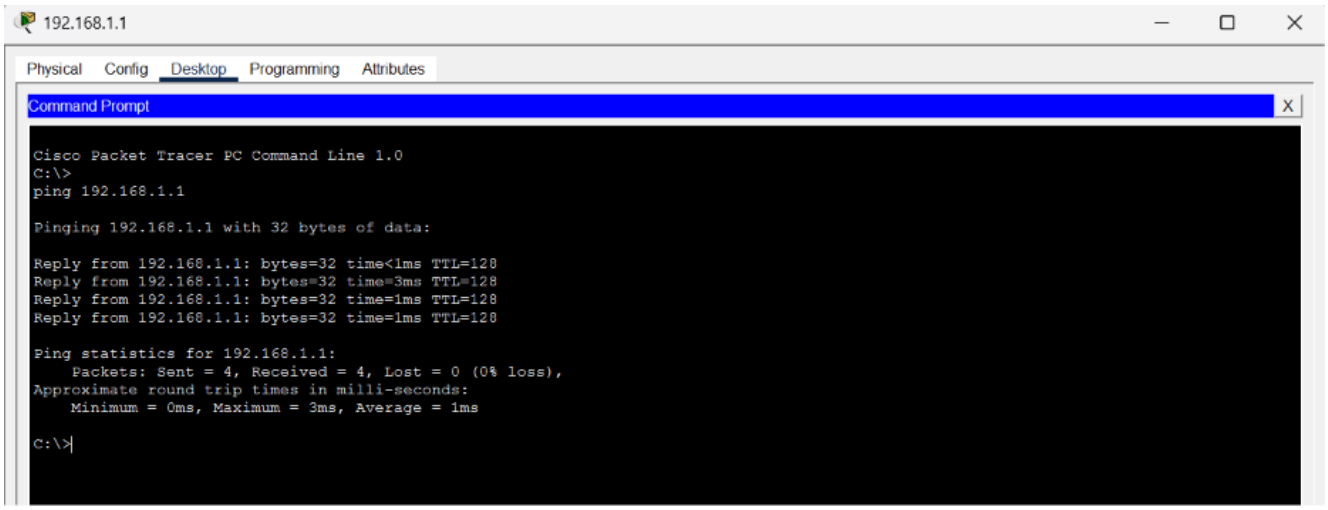
Study the concept of VLAN using packet tracer.

Practical Assignment #05:

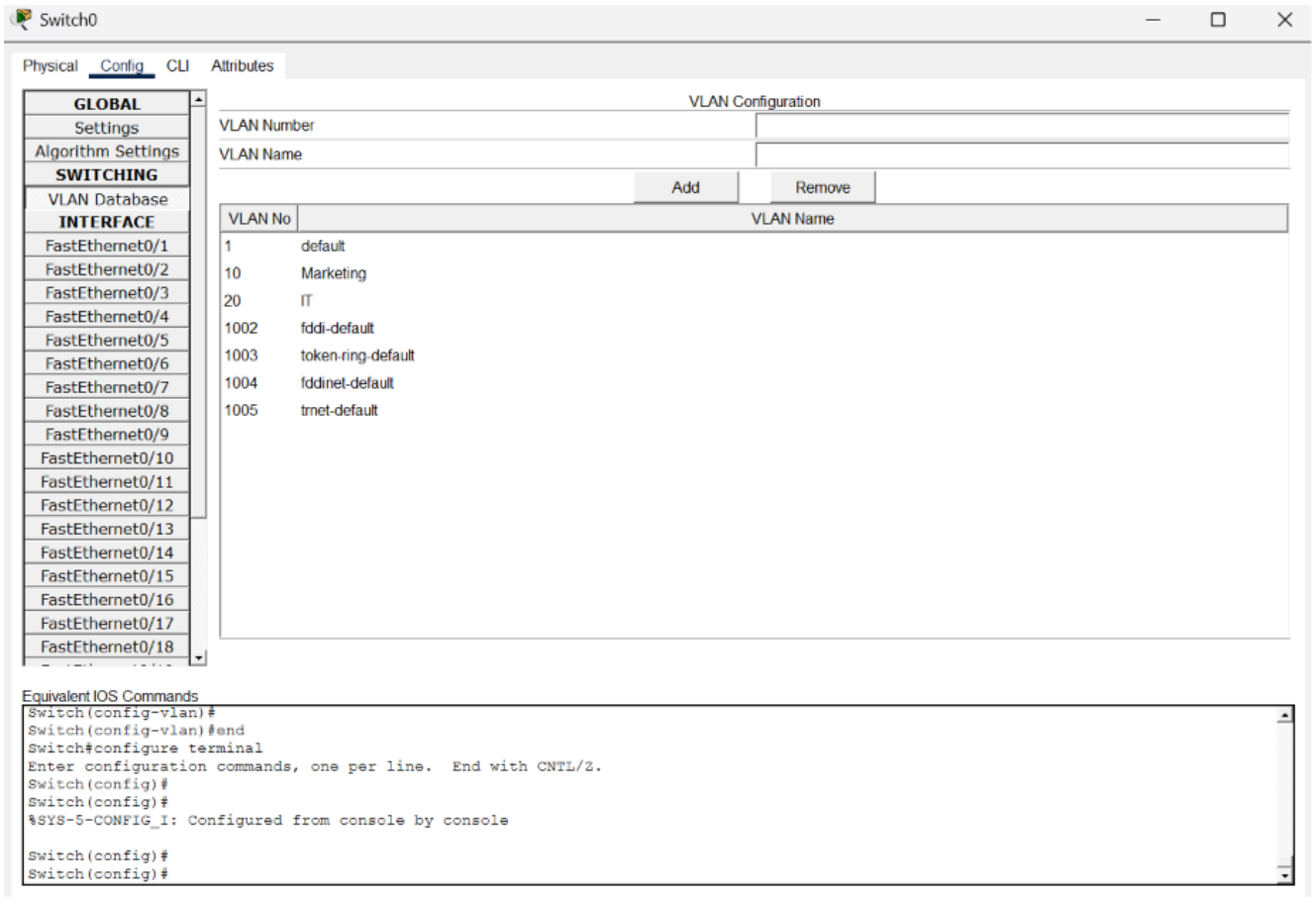
1. **Implement the different network structures in VLAN and VLAN trunking. Also check connectivity between them using ping command or PDU utility.**

1 VLAN :-

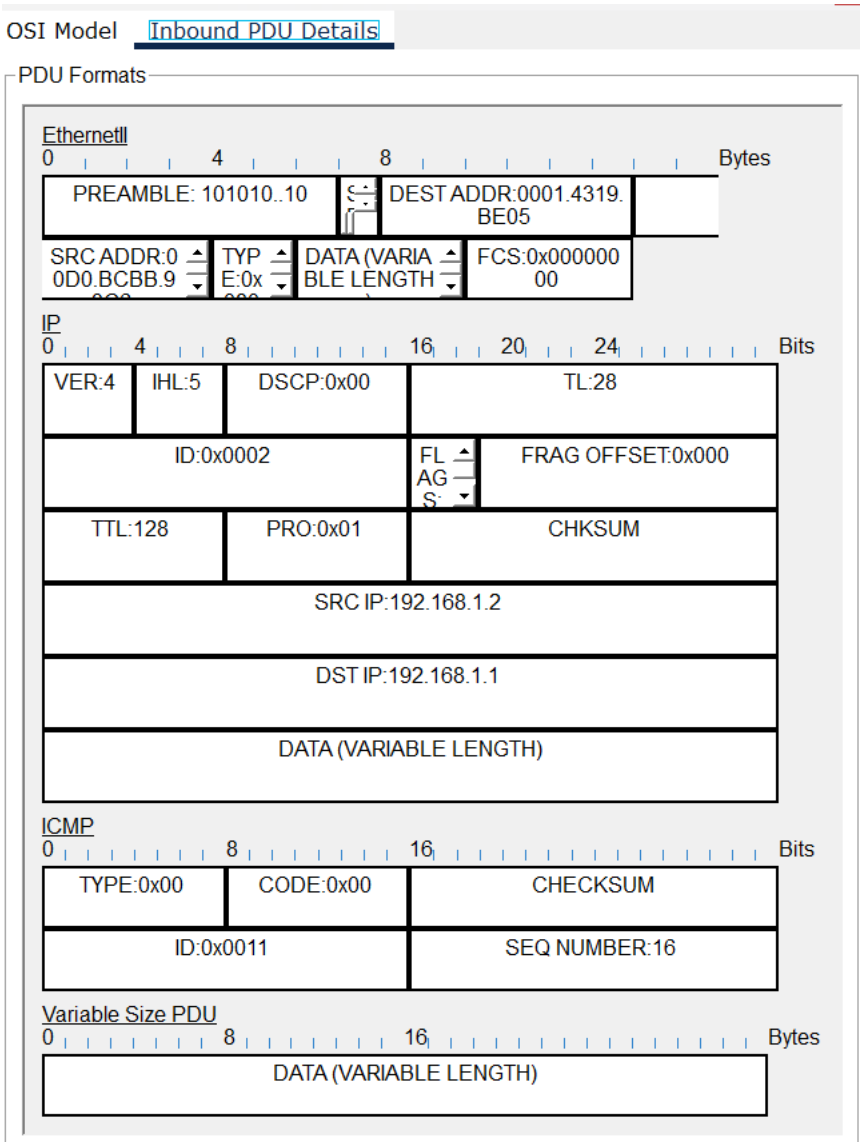




Date: 09/07/2026



Date: 09/07/2026



Steps :-

- Take 4 pc and assign Ip address and label to each pc.
- Take 2 Switch and Make 2 Lan with 2 pc each.
- Now there is local area network but We need to implement Virtual Lan database for implement Virtual Lan we need to configure Virtual Lan database in configure Section in switch.
- Make 2 New Virtual Lan configuration with unique Virtual Lan number and name.
- Now check the Port where the pc is connected and open that port configuration

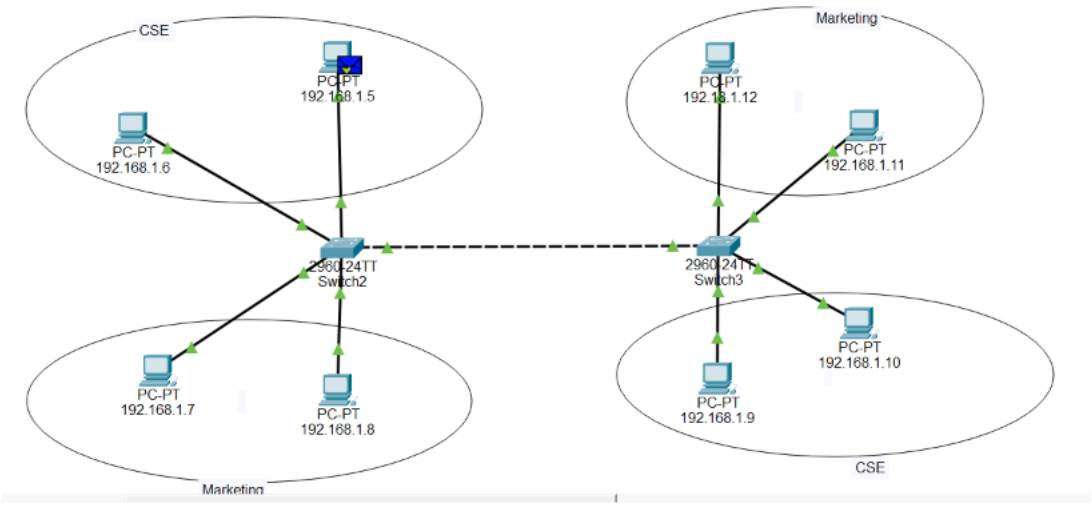
Date: 09/07/2026

in switch and set Virtual Lan to access mode and select your Virtual Lan database name in dropdown.

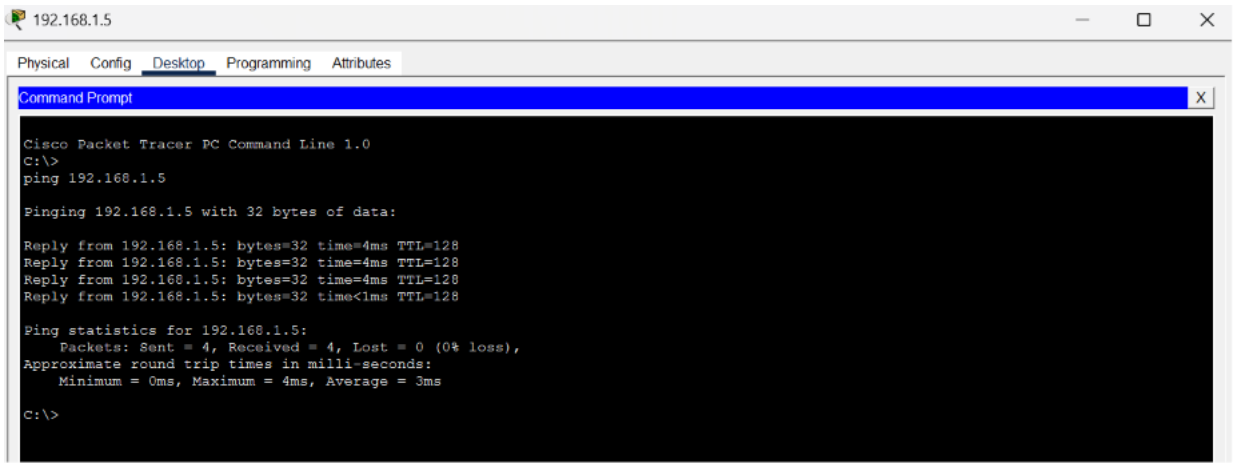
• Now we need to connect both Switch but here we need to give Virtual Lan mode to trunk because there are number different Virtual Lan signal travel threw it.

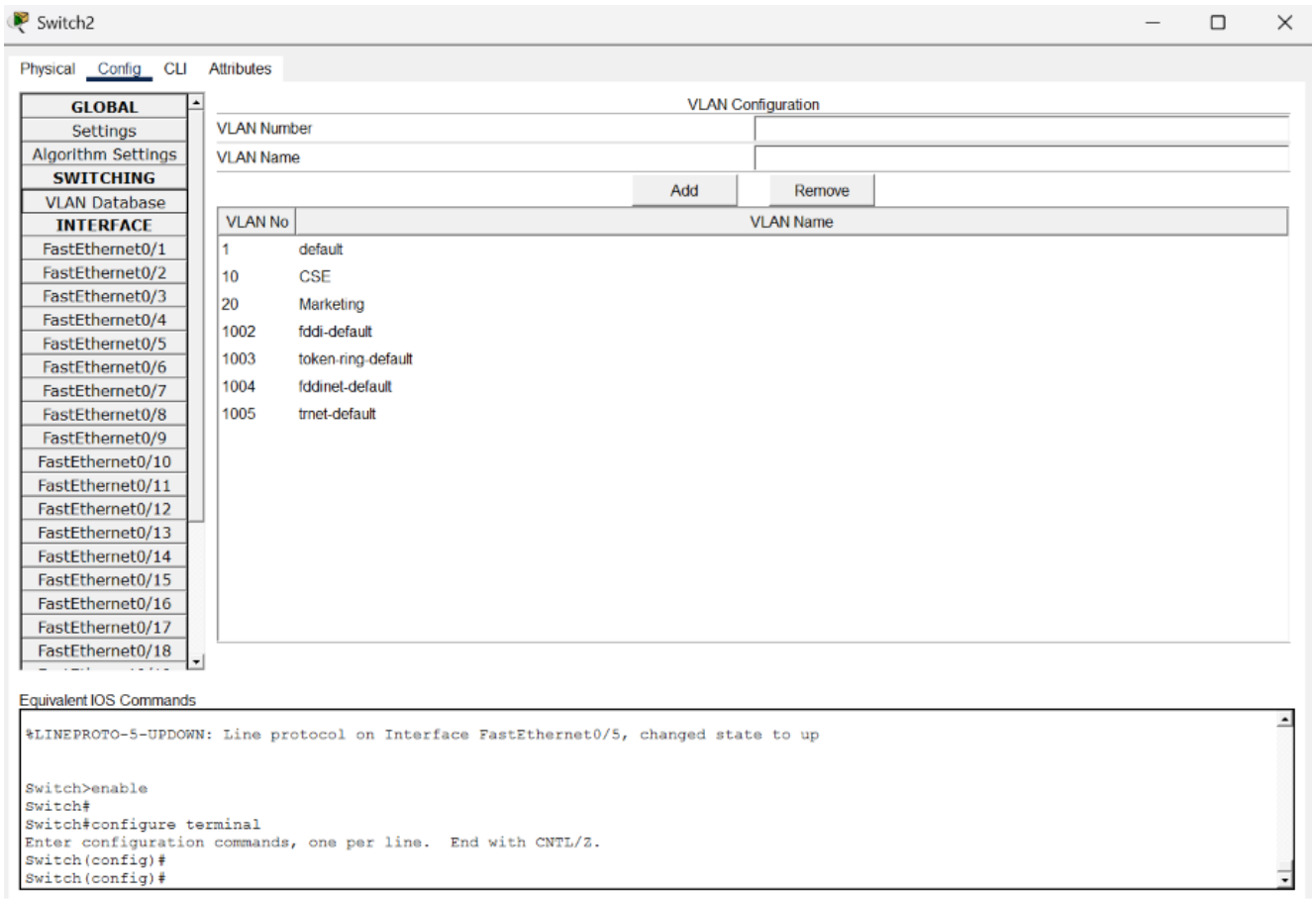
• Now you configure two Virtual Lan in different local area network.

2 VLAN:-

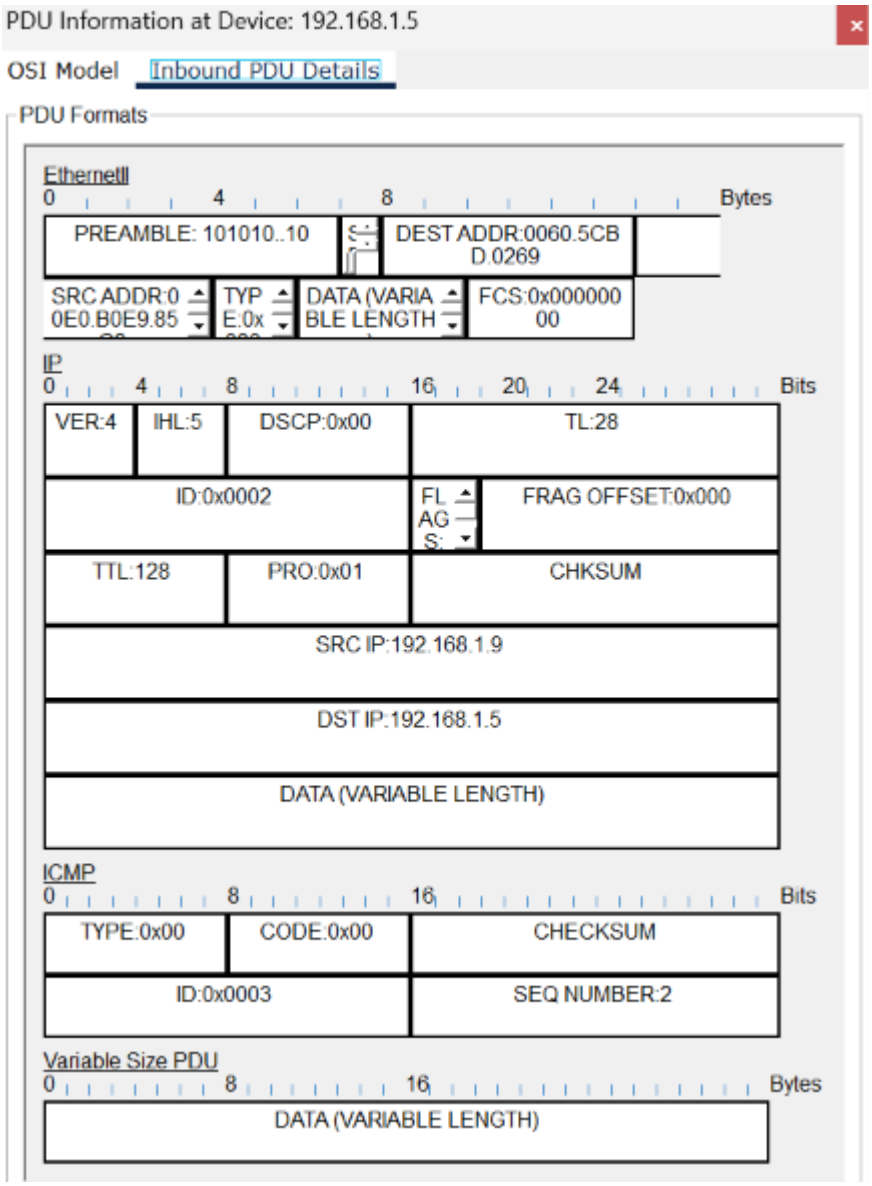


Date: 09/07/2026

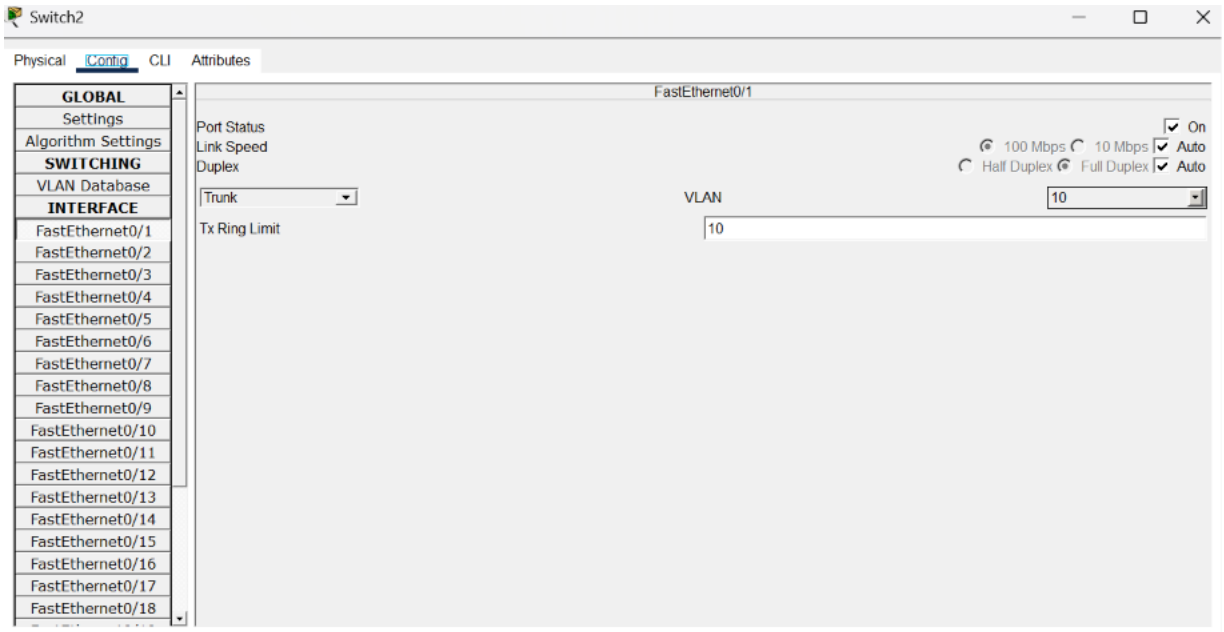




Date: 09/07/2026



Date: 09/07/2026

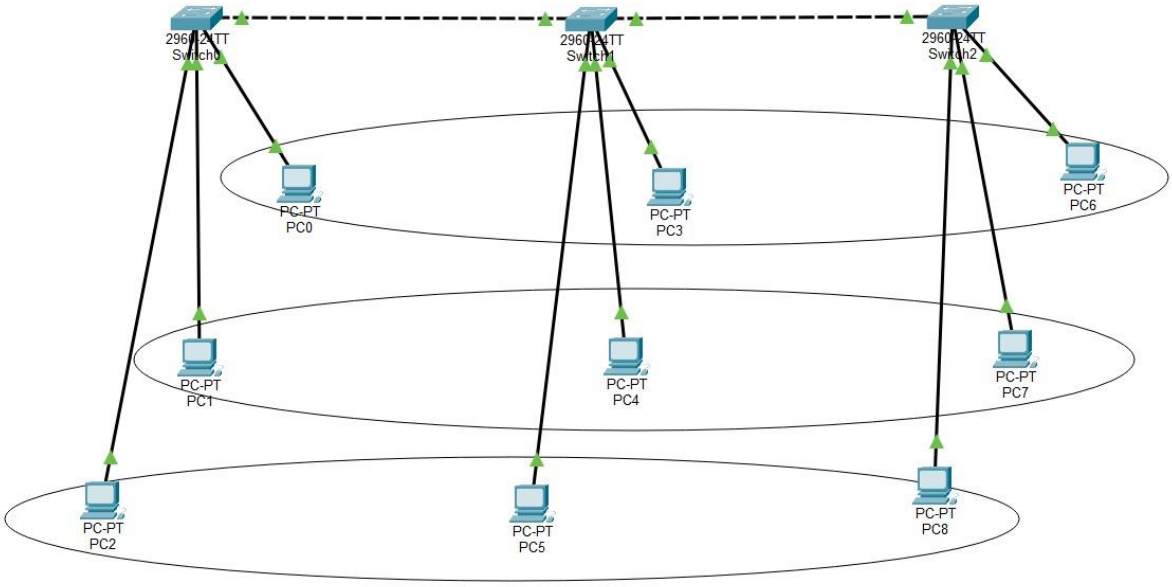


Steps :-

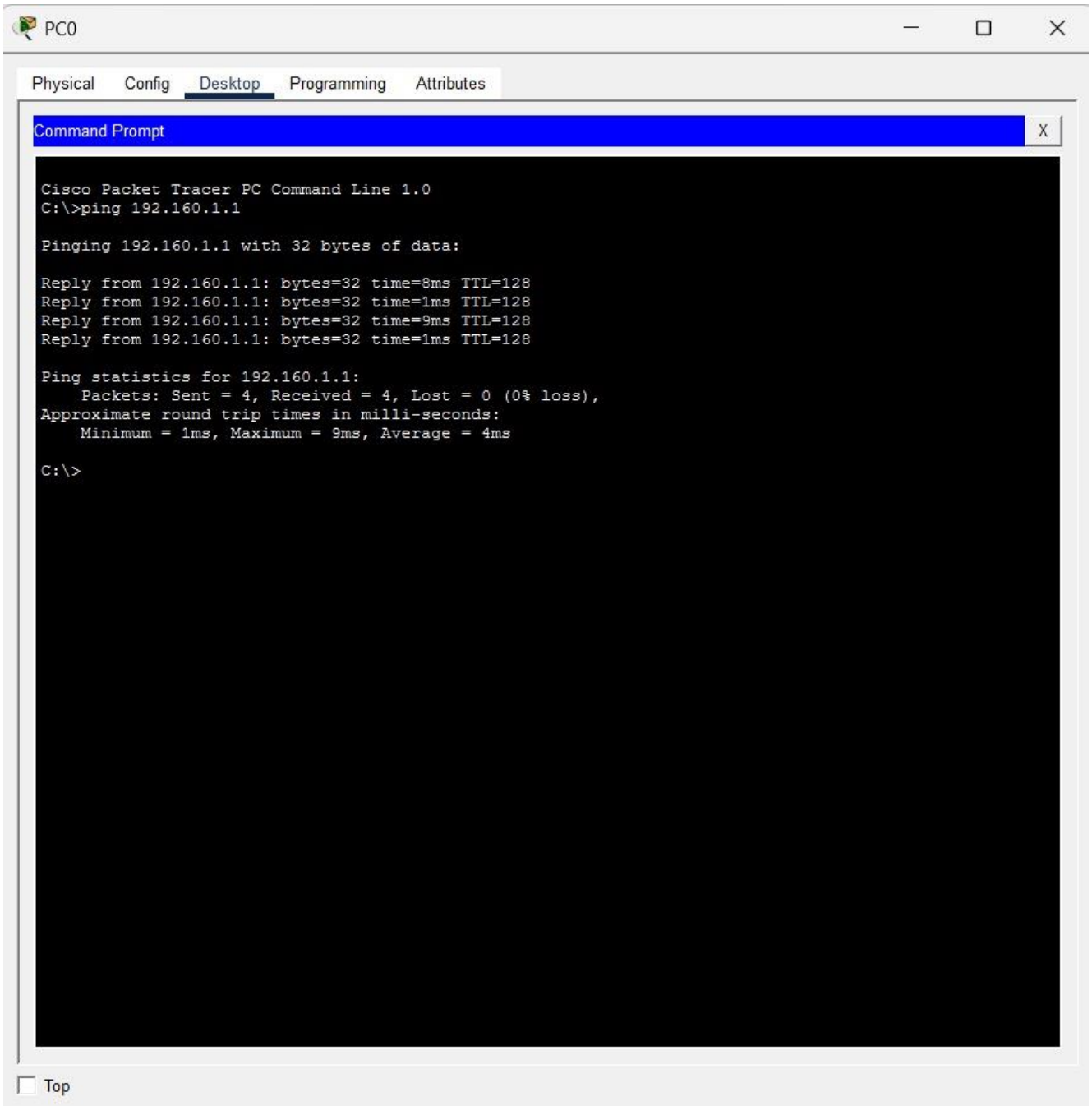
- Take 8 pc and assign Ip address and label to each pc.
- Take 2 Switch and Make 2 Lan with 4 pc each.
- Now there is local area network but We need to implement Virtual Lan database for implement Virtual Lan we need to configure Virtual Lan database in configure Section in switch.
- Make 2 New Virtual Lan configuration with unique Virtual Lan number and name.
- Now check the Port where the pc is connected and open that port configuration in switch and set Virtual Lan to access mode and select your Virtual Lan database name in dropdown.
- Now we need to connect both Switch but here we need to give Virtual Lan mode to trunk because there are number different Virtual Lan signal travel threw it.
- Now you configure two Virtual Lan in different local area network.

Date: 09/07/2026

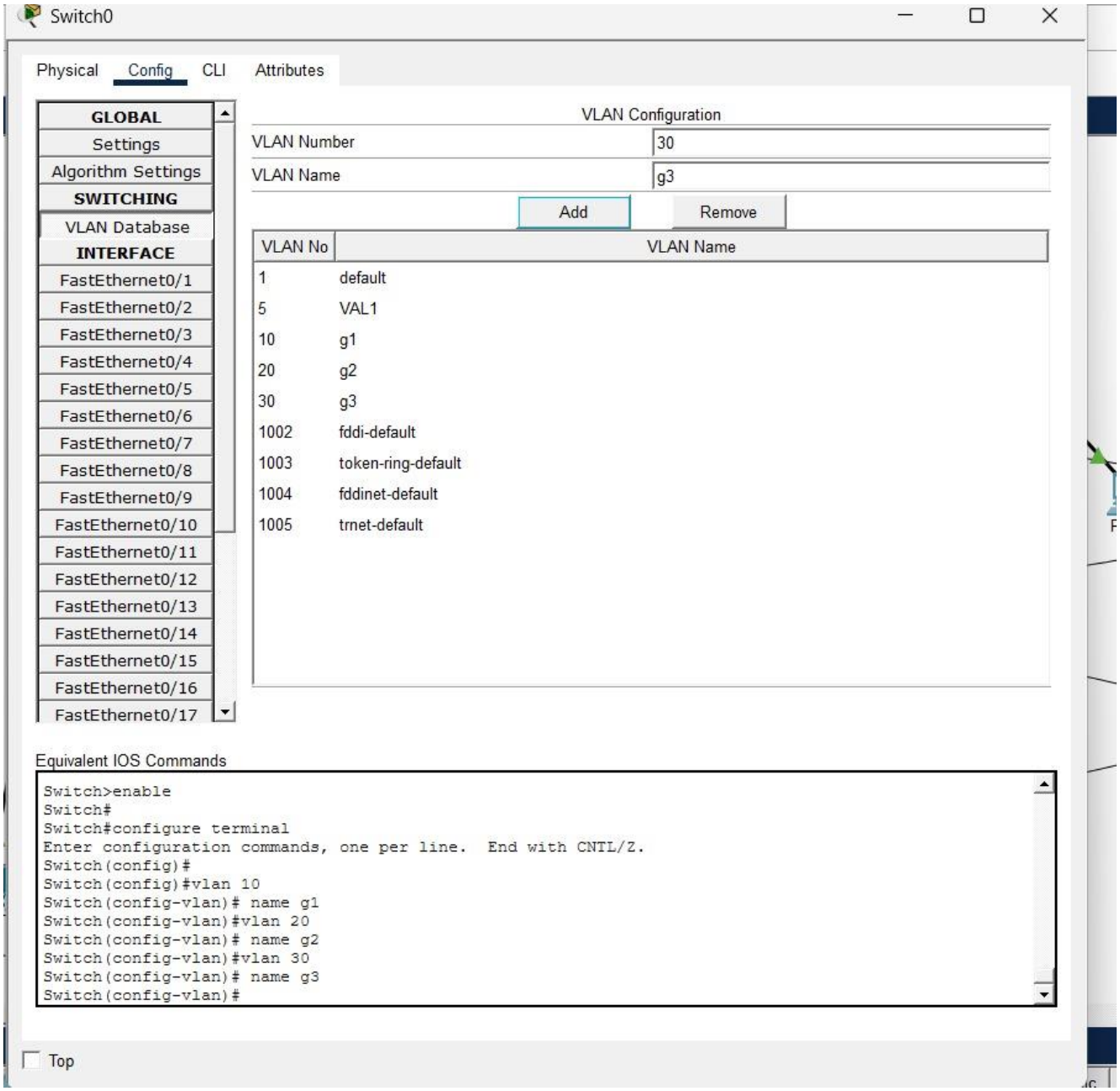
3 VLAN :-



Date: 09/07/2026



Date: 09/07/2026



Steps:

- Take 9 PCs and assign IP address and label to each PC.
- Use 3 switches and make a LAN where each switch connects to 3 PCs.
- Now there is local area network but we need to implement Virtual LAN database.
- For implementing Virtual LAN, configure VLAN database in the Configure section

Date: 09/07/2026

of each switch.

- Make 3 new Virtual LANs with unique VLAN numbers and names (ex. 10: g1, 20: g2 , 30:g3).
- Check the port where the PC is connected, open the Port Configuration in the switch, set port mode to Access, and select the correct VLAN from dropdown.
- Connect all switches together.
- On the ports between switches, set mode to Trunk because multiple VLAN signals will travel through it.
- Now you have 3 Virtual LANs configured across different LANs.
- Verify connectivity using PDU or ping. Devices in same VLAN should communicate successfully; different VLANs won't communicate without routing.